

An intelligent web-based platform for processing Digital Calibration Certificates utilizing the GEMIMEG tool and advanced AI capabilities

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ABSTRACT

With the growing emphasis on digitalization in the metrology industry, significant efforts have been made to improve the process of standardization and transition from traditional calibration certificates to digital calibration certificates (DCCs). This shift is driven by the rapid pace of digital transformation in metrology, necessitating that National Metrology Institutes (NMIs) and industrial standard laboratories adopt robust software protocols for data storage, handling, and secure transfer. Furthermore, they must establish effective procedures for generating and issuing DCCs.

This work introduces an innovative web-based platform designed to facilitate the issuance of DCCs, particularly for a basic calibration laboratory. Digital certificates also simplify manual and automated interaction during the measurement process, allowing for the integration of machine learning and artificial intelligence (AI) techniques to analyze calibration data in Internet of Things (IoT) systems. Our web-based graphical user interface (GUI) oversees all DCC-related processes, from service requests to final certificate issuance. The platform utilizes the GEMIMEG tool, developed by PTB, to generate DCCs through various input masks, providing a streamlined method for producing administrative data. By employing XSLT stylesheets, human-readable documents can be generated from the DCCs.

1. Introduction

DCCs offer several key advantages over traditional certificates, such as instant data download to instruments without human intervention, thanks to their machine-readable format. This automation enables quick verification of certificate availability, validity, and content, reducing human error in data entry. As metrology devices become more complex and calibration certificates contain more information, the need for digital solutions becomes apparent.

To demonstrate the practical application of our platform, DCCs are created for digital load cells, one of the most common electronic devices in metrology. The environment is designed to be secure, with all data processed and stored in the cloud. Our future plans include extending this solution to a broader range of metrological areas and services.

Currently, there is a strong push to transition from conventional paper-based calibration certificates to digital calibration certificates (DCC). Leading this movement is Germany's Physikalisch-Technische Bundesanstalt (PTB), which has allocated funds for research initiatives aimed at standardizing the format of DCC.

A. Digital calibration certifications, DCC

Digital calibration certificates have multiple functions, but their key role is to support true digitalization, which uses digital technologies to transform business models and create new opportunities with enhanced value. Developing a system to exchange digital calibration certificates will allow customers to receive and analyze these certificates more quickly, cutting down on the time and costs related to metrological

confirmation processes while also reducing the risk of human error and its consequences. This system also makes it easier to exchange calibration results. The straightforward process of collecting and analyzing information from a sequence of calibrations allows for more comprehensive examination of the data obtained during the calibration.

The Digital Calibration Certificate (DCC) goes beyond simply transferring the contents of an analogue calibration certificate, where measurements are printed on paper with distinct characteristics, qualifying it as a document. The DCC does more than merely convert paper into a digital format like a PDF; it is designed to be both human-readable and machine-readable. This is achieved by storing data in a structured format. As a result, the DCC can store a large volume of information in a more structured manner than previous methods allowed [1,2].

The development of a new type of calibration certificate is increasingly crucial for various reasons. These include the growing connectivity of measurement devices and the rising use of IoT systems with multiple measurement sensors. Additionally, the emergence of measurement-intensive applications and devices, such as autonomous cars, robots, and remote surgical operations, further underscores this need. These applications contain integrated measurement components that require calibration and automated processing of calibration data [3–6]. The dependence on manual intervention to analyze and verify calibration certificate contents presents significant limitations for these new technologies [3].

B. Digital load cells

A load cell is a device designed to measure force, weight, or pressure,



Fig. 1. Examples of practical digital load cells under study.

typically using digital technology. Inside a load-bearing structure, such as a bending beam, four resistive strain gauges are installed. A Wheatstone bridge setup allows the mechanical stress exerted on the beam to be converted into electrical signals via the four strain gauges. To ensure accuracy, the strain gauges are calibrated using a test scale with known forces.

Digital load cells are equipped with digital technology for measuring force, weight, or pressure. These devices typically feature four resistive strain gauges mounted within a bending beam or similar structure. The Wheatstone bridge design transmits the force onto these gauges, converting the stress into electrical signals. Calibration is done by applying known forces to ensure the accuracy of the measurement.

Digital load cells are versatile components in force and pressure measurement systems, applicable to a wide range of operations, regardless of the size, weight, or cost of the final product. They are suitable for compact applications, particularly where there's limited space, as they don't require bulky digital displays or external amplifier boards. These digital load cells have built-in signal processing circuits, providing a fully processed output without the need for additional processing, filtering, or averaging. The processing is all done within the load cell's internal electronics, ready to be integrated into different types of systems [7].

Load cells, which are useful in various applications, can be created by combining a range of mounting hardware and electronics. These weight sensors come in different sizes and designs to fit even the most challenging installation needs. They are designed for use in applications that require small to large capacity load cells. The ease of collecting and analyzing information from a series of calibrations allows for a more comprehensive assessment of the data obtained during calibration.

Load cells are used in various applications, as shown in Fig. 1. Examples include truck scales, railway scales, supermarket scales, bench scales, and heavy-duty scales. These are just a few instances where load cells are used in weighing equipment. In process weighing, load cells find applications in conveyor scales, high-speed check weighers, multi-head packaging machines, silo/tank weighing systems, and machines for bagging and bottle filling. In medical devices, load cells are used in patient lifts, hospital beds, baby incubators, infusion pumps, baby scales, and exercise machines. In the field of test and measurement, load cells are used in hand tools, dynamometers, press controls, material testing machines, haptic devices, and general testing setups. For process automation control, load cells are utilized in tablet presses, printing equipment, inventory control systems, electronic assembly tools, and well pumps [7,8].

Given the critical applications of load cells, it's essential to transition to digital technology in this field to align with the rapidly evolving tech landscape. To advance in this direction and meet the future demands of society, including digitalization and the requirements of Industry 4.0 and IoT, we need to create appropriate plans for issuing calibration certificates in a digital, machine-readable format. This will make load cell metrology compatible with emerging technologies and future-proof for anticipated needs. Therefore, we propose implementing and testing a new web-based environment for load cell calibration services [8].

C. GEMIMEG-II project [9].

The GEMIMEG-II project is a flagship initiative funded by the German Ministry of Economic Affairs and Energy as part of its "Digital Technologies for the Economy" program. It aims to digitize metrology—a discipline that deals with various measurements—using a multidisciplinary approach involving industry, research institutes, and universities. To facilitate the adoption of digital technologies by the economy and society, the project develops its concepts through small and medium-sized industrial demonstrators known as "real beds."

A reliable and standardized infrastructure is essential for the widespread acceptance of digital products in industrial settings. Thus, GEMIMEG-II is creating a digital platform to streamline measurement processes. A key element of this project is the digital calibration certificate (DCC), which establishes uniform methods and procedures for the robust, machine-readable transfer of metrological data. The project's developments are tested, refined, and extended for industrial use by various partners both within and outside PTB. The GEMIMEG-II project is supported by the German Ministry of Economic Affairs and Energy.

D. Problem statement, objective and paper structure

Due to the efforts of PTB (Germany), the GEMIMEG-II research project yielded the launch of an online tool, also called "GEMIMEG-II," several months ago. This tool allows users to generate digital calibration certificates (DCCs) with its latest version. PTB is currently working on updating the file structure for these DCCs. A release candidate (DCC schema 3.3.0-rc.1) has been published as part of the ongoing development of the DCC schema.

Our primary goal is to streamline digital communication between instrument owners (or "customers") and calibration provider staff, making it easier for customers to obtain digital calibration certificates (DCCs) for their devices. To achieve this, we've developed a modern website equipped with various AI tools to simplify these interactions. The second objective focuses on helping primary calibration labs with less developed digital infrastructure to manage the process of collecting all necessary data for calibration procedures. Additionally, they can utilize the online tool "GEMIMEG-II" to create DCCs in accordance with the latest industry standards. Our third objective is to provide a secure electronic storage system with approximately 50 GB of space, which can safely house all issued certificates. The following sections outline related work in section 2. Section 3 discusses the design and implementation of our proposed web-based environment in detail. Section 4 presents the evaluation of the system, and the summary is provided in section 5.

2. Related work

In our earlier work [10,11], we focused on establishing a fundamental digital infrastructure for calibration laboratories to facilitate the production of Digital Calibration Certificates (DCCs). Over the past few years, several other approaches have been proposed to streamline the DCC production process. For example, in Ref. [12], a web-based graphical user interface (GUI) was designed to create, visualize, and operate NAWI-DCC files. This interactive application, built with Shiny, provides an easy way to generate human-readable PDF versions of calibration certificates and convert indicator readings into mass

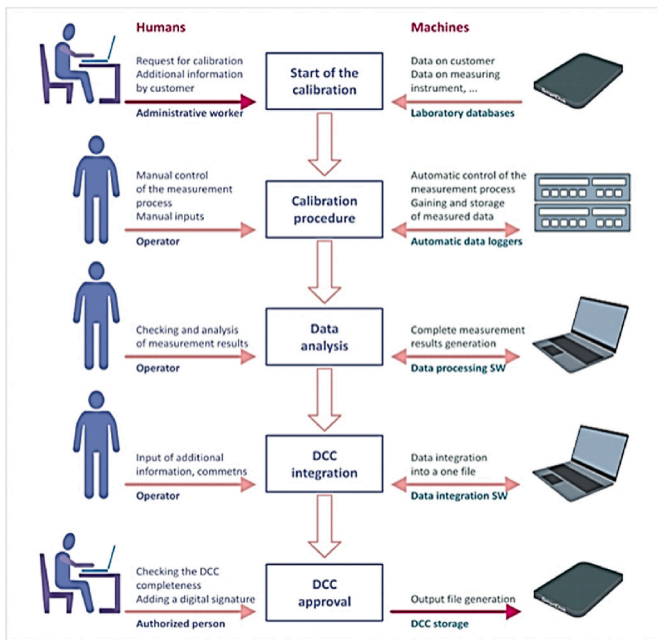


Fig. 2. Different processes of generating DCCs [14].

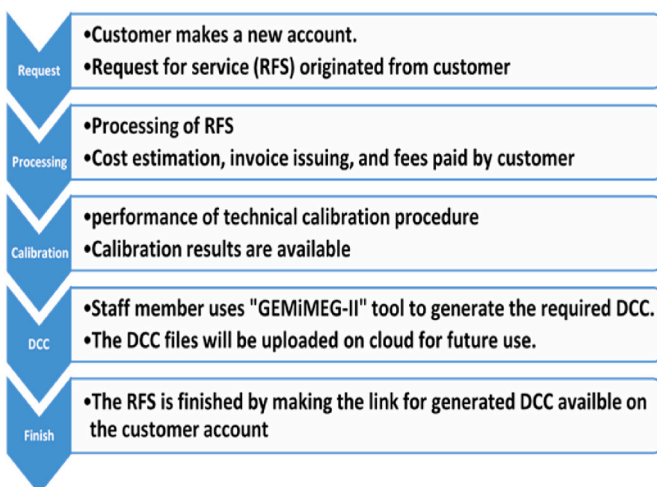


Fig. 3. Process flow diagram for the interaction between customers and staff.

measurement results with corresponding uncertainties. Another example, in Ref. [13], describes a prototype web application developed in the programming language Links, designed to store and display DCCs using a relational database. In more recent work [14], the “Monte Carlo Method” was proposed to examine various stages in the production of DCCs, as shown in Fig. 2.

3. Proposed system

In the planning phase for the proposed environment, the process flow diagram depicted in Fig. 2 is taken into account.” The proposed web-based platform enables secure communication between customers and staff at the calibration lab. The site has a sleek logo and uses carefully chosen images from Freepik [15] to avoid copyright complications. The design emphasizes user experience, ensuring easy navigation throughout the main interface. The code is written in a structured and readable way, allowing other developers to build upon it without difficulty. The system’s primary users are customers or calibration lab staff.

From the customer’s perspective, the initial step is to log into the website’s home page (Figs. 3–5) {<https://dcc.readyom.com/>} [16] and create an account, providing all necessary company details. The system automatically generates a unique username and password for the customer to ensure security. Once the account is set up, customers can submit a Digital Calibration Request –DCR- (Request for Service “RFS”). The DCR contains all the information required for the calibration of a specific device (Device Under Test “DUT”). Existing customers who log into the system can track the status of their request, which can be “Open,” “Under Process,” or “Finished”. As an optional track, an old DCC for the DUT can be uploaded on the website. All the basic information can be extracted automatically from such uploaded file for future use by lab staff. After the calibration process is completed by the lab staff, the customer can obtain the issued digital calibration certificate (DCC) through the online system.

Calibration lab staff can access the system by logging in with their email as the username and a password, which is automatically generated by the system when the staff member creates an account (refer to Fig. 6). A designated staff member must review the submitted Request for Service (RFS) and the Device Under Test (DUT) to initiate the technical steps of the calibration process. Moreover, if an old DCC was uploaded by customer, their extracted basic information can be used during filling-in different fields required by the “GEMIMEG-II” tool. After calibration procedures have been finished correctly, calibration results are stored for future reference. A simple script (using Python or Selenium) can be used to automatically fill in all the required fields of the “GEMIMEG-II” tool. Finally, the tool issues the relevant Digital Calibration Certificates (DCCs).

The Digital Calibration Certificate (DCC) is generated as an “.xml” file, but it can also be converted into a “.pdf” format for easier human review. You can find an example of a DCC for a load cell device in Fig. 7. Once the DCC is issued, whether in “.xml” or “.pdf” format, a staff member uploads it to the corresponding open RFS (Request for Service) account. After its uploaded, customers can see a download icon next to their respective RFS, allowing them to download the certificate. All such activities are managed through the front-end of a web-based platform. The front-end was developed using various programming languages and frameworks, including Bootstrap v4.3.1, Font Awesome, Laravel 10, and PHP 8.1. All data, whether submitted by customers or staff members, is stored in the back-end database, which is built using MySQL.

Incorporation of different AI tools can significantly enhance the user experience, streamline operations, and provide added value to clients. Here are several innovative AI tools and features that can be integrated into the website: One of the most impactful AI features is a virtual assistant or chatbot, which can provide real-time assistance to website visitors. This tool can answer frequently asked questions, guide users through the process of requesting calibration services, and even help customers track the status of their calibration certificates. The chatbot can use natural language processing (NLP) to understand user queries and provide appropriate responses, offering a convenient and interactive experience without needing human intervention. Another valuable AI tool is a recommendation system. By analyzing user interactions, such as the types of equipment frequently calibrated or the services commonly requested, the website can offer personalized recommendations. This could help clients identify additional services or suggest calibration frequencies based on industry best practices or regulatory requirements. The recommendation system can also offer relevant content, like technical articles or industry news, tailored to users’ interests. AI-driven data analytics can be employed to monitor equipment calibration trends and predict when recalibration might be necessary. This can be displayed on the website as a value-added service, allowing users to anticipate future calibration needs. Predictive analytics can help organizations optimize their maintenance schedules and reduce downtime, enhancing customer satisfaction. For an organization issuing calibration certificates, document management is critical. AI-powered document processing can automate certificate issuance and

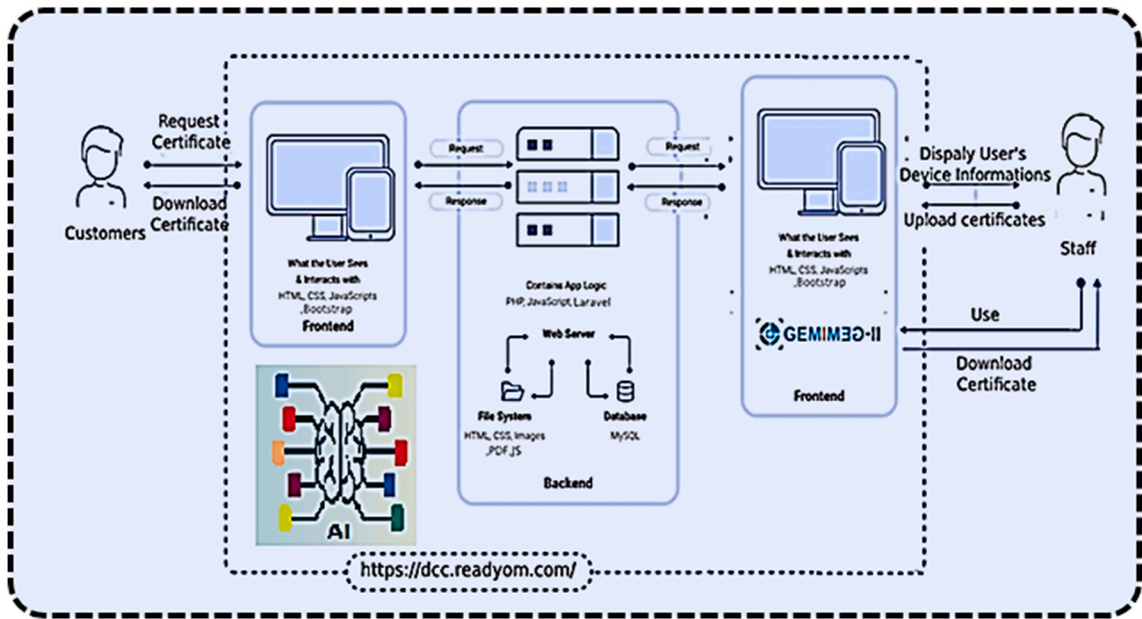


Fig. 4. The architecture of the proposed web-based environment equipped with AI tools.

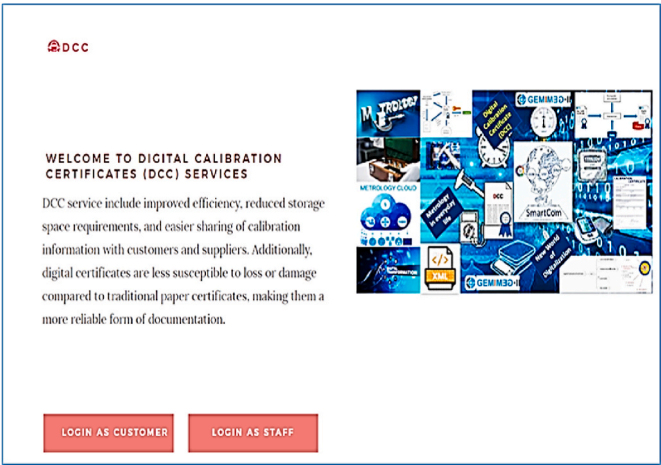


Fig. 5. The first page (GUI) of our proposed web-based environment system.

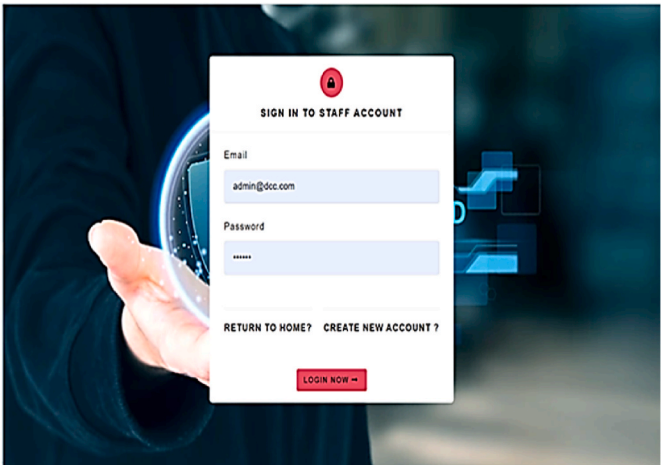


Fig. 6. Page of signing-in for staff members.



Fig. 7. Example of a DCC generated by GEMIMEG tool for a load cell.

archival. By implementing machine learning algorithms, the website can streamline the process of generating calibration certificates, ensuring accuracy and reducing manual labor. This automation can extend to searching and retrieving certificates, allowing customers to access their calibration history easily.

Another AI application that can be integrated is computer vision. If the organization uses visual inspection during calibration, a computer vision tool can analyze images or videos to detect anomalies or deviations from standards. This technology can help maintain high-quality calibration services and can be used to provide visual evidence to customers, increasing transparency. Finally, AI-based security features can be added to enhance the website's safety and protect sensitive data. Tools like anomaly detection can monitor website traffic and user behavior, identifying unusual patterns that might indicate a security threat. This enhances customer trust and ensures compliance with industry regulations regarding data protection.

By integrating these AI tools and features, a website for an organization issuing calibration certificates for industrial equipment can become more user-friendly, efficient, and secure, offering clients a

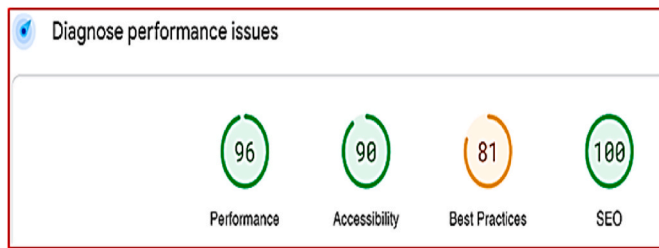


Fig. 8. Performance metrics of the proposed website.

seamless and high-quality experience.

4. Proposed system assessment

To evaluate a website created for issuing digital calibration certificates, several online tools can be used to gauge its performance, user engagement, functionality, security, and other essential aspects. Google PageSpeed Insights is one such tool, providing insights into a website's speed and performance on both mobile and desktop platforms. It also offers suggestions for improving loading times. The proposed website was evaluated using this tool {<https://pagespeed.web.dev/?hl=ar>} [17]. Fig. 8 shows the results of the performance metrics. Four metrics were used for the assessment:

Performance: This measures how quickly the website loads and responds to user interactions. A high performance score of 96 % indicates excellent responsiveness.

Accessibility: This metric assesses how easily users with disabilities can access and use the website. The accessibility score was 90 %, which is quite good.

Best Practices: This measures how well the website adheres to industry-standard best practices in web development. An 81 % score indicates a generally good level of compliance.

SEO (Search Engine Optimization): This metric evaluates how well the website is optimized for search engines like Google. A perfect score of 100 % suggests that the site is highly optimized.

This evaluation is considered self-assessment. Additionally, the proposed environment can also be evaluated from a user's perspective, which is often referred to as "opinion mining." This approach requires feedback from many users for accurate results. The team plans to explore this method in future work.

5. Summary

In this study, our primary aim is to make it easier for customers and calibration laboratory staff to obtain the most recent version of the Digital Calibration Certificate (DCC). To achieve this, we designed, developed, and implemented a user-friendly web-based environment. This platform has been evaluated and shows promising results in four key areas: performance, accessibility, best practices, and search engine optimization.

In the future, additional digital services, such as DCC verification, could be integrated into this environment. Furthermore, user opinion mining might be employed as an additional method for evaluation to gain deeper insights into user feedback and improve the overall experience. Further work for investigation of the workability of the built

web-based environment is planned. Extending the implementation to include other metrological fields/services; such as thermometry, mass metrology, optical metrology, electrical metrology and length metrology are targeted as extension of this work. Further demonstrations and continuous is of essential need to move further as of the target plan.

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References

- [1] <https://dccwiki.ptb.de/en/home> {Accessed 20 April 2024, 11 AM }.
- [2] M.S. Gadelrab, R.A. Abouhagail, Towards a new generation of digital calibration certificate: analysis and survey, *Measurement* 181 (2021) 109611.
- [3] <https://www.gemimeg.ptb.de/gemimeg-tool>.
- [4] <https://dccwiki.ptb.de/en/Tools/Excel-Tool> { Accessed 20 April 2024, 11 AM }.
- [5] <https://dccwiki.ptb.de/en/downloads> { Accessed 20 April 2024, 11 AM }.
- [6] A.A. Al-Dulaimi, S.M. Al-Mashhadany, M.A. Al-Zubaidy, A.A. Al-Dulaimi, Websites optimization metrics: a systematic literature review, *J. Prospec. Res.* 24 (2) (2024) 42–53.
- [7] A. Cunillera, N. Bešinović, R.M. Lentink, N. van Oort, R.M. Goverde, A literature review on train motion model calibration, *IEEE Trans. Intell. Transport. Syst.* 24 (4) (2023) 3660–3677.
- [8] N.S. Saedin, M.F.A. Muttalib, M.F. Jusoh, Performance evaluation of bar load cell sensing system for soil moisture measurement, *J. Phys. Conf.* 2550 (1) (2023, August) 012013. IOP Publishing.
- [9] <https://www.ptb.de/cms/en/ptb/fachabteilungen/abt5/fb-53/forschungsvorhaben-53/gemimeg-ii.html> { Accessed 20 April 2024, 11 AM }.
- [10] K.M. Ahmed, et al., Smart infrastructure for developing digital calibration certificates at SASO-NMCC, in: *Proceedings of M4DConf2022, TC6/IMEKO, PTB, 2022*.
- [11] K.M. Ahmed, et al., A web-based environment for processing digital calibration certificates by employing GEMIMEG-Tool, in: *NCSLI Workshop & Symposium | July 8 - 12, 2023, Orlando, Florida*.
- [12] Crithian Paredes, et al., A digital calibration certificate prototype for non-automatic weighing instruments using the open source R package masscor, in: *NCSLI Workshop and Symposium, Nourishing Metrology Professionals, August 20-24, 2022, Grapevine, TX*.
- [13] Vashit Galpin, et al., Tracking and viewing modifications in digital calibration certificates, *ACTA IMEKO* 12 (1) (March 2023) 1–7.
- [14] G. Sammarah, M. Halaj, L. Bartalský, J. Palenčár, Monte Carlo simulation in a digital calibration certificate, in: *EAI International Conference on Automation and Control in Theory and Practice, Springer Nature Switzerland, Cham, 2023 February*, pp. 345–357.
- [15] <https://www.freepik.com/> { Accessed 20 April 2024, 11 AM }.
- [16] <https://dcc.readyom.com/> { Accessed 20 April 2024, 11 AM }.
- [17] <https://pagespeed.web.dev/> { Accessed 20 April 2024, 11 AM }.

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